COL761/COL7361/AIL7026
Data Mining
Assignment 2: Q1 k-means Clustering Report

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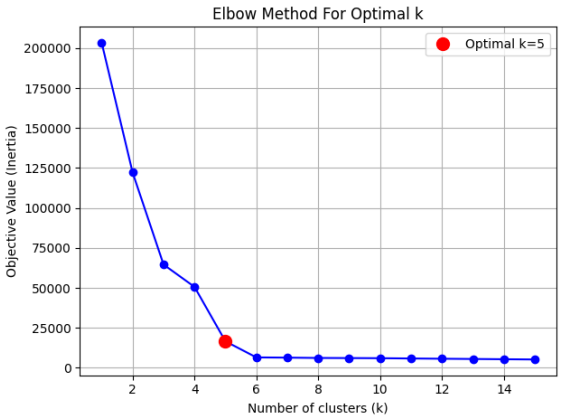
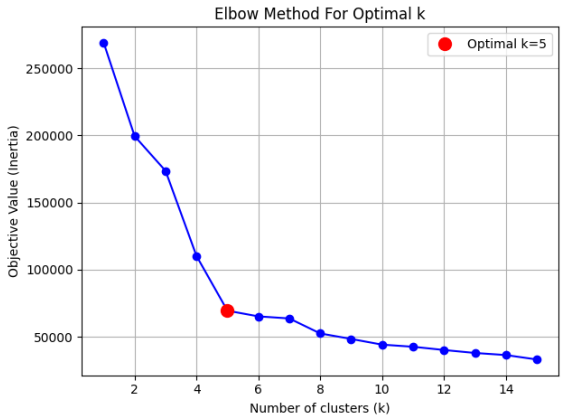
March 16, 2026

1 (Computational Task Results)

The k-means algorithm was implemented for $k \in \{1, \ldots, 15\}$. The objective value (Inertia) were recorded for two distinct datasets. In both cases, the optimal number of clusters was determined to be $k = 5$.



(a) Dataset 1: Elbow Point Graph



(b) Dataset 2: Elbow Point Graph

1.1 (a) Reasoning for the Selected Value of k

Based on the empirical evidence, $k = 5$ is the optimal choice for both datasets.

- **Dataset 1:** Shows a sharp "elbow" at $k = 5$ where the inertia drops by nearly 90% from the $k = 1$ baseline.
- **Dataset 2:** Shows a similar inertia plot as in Dataset 1, even with more number of columns.

1.2 (b) Structural and Statistical Assumptions

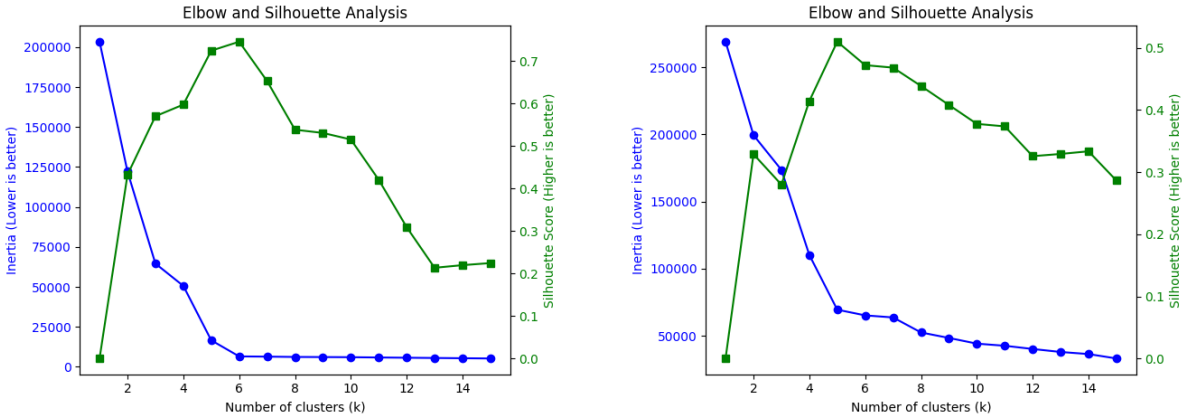
The k-means objective, $\min \sum_{j=1}^{k} \sum_{x_i \in C_j} \|x_i - \mu_j\|^2$, relies on:

- **Structural Assumptions:** Clusters are spherical (isotropic) and have similar spatial extents and densities. It effectively assumes features are independent and have equal variance.
- **Statistical Assumptions:** k-means is a restricted case of the Expected-Maximization algorithm for Gaussian Mixture Models (GMM), assuming each cluster is generated from a Gaussian distribution with an identify co-variance matrix ($\Sigma = \sigma^2 I$).

1.3 (c) Methodological Improvements

To move beyond the subjective "visual elbow" heuristic, we implemented ***Silhouette Analysis*** as a secondary validation principle.

- **Validation:** By checking if the Silhouette peak aligns with the Elbow point:
  - For Dataset-1, the inertia drops by nearly 90% from the $k = 1$ baseline. While the Silhouette score peaks at $k = 6$ (≈ 0.75), the score at $k = 5$ is extremely high (≈ 0.73). This also proves why we selected $k = 5$, as the additional cluster provides negligible variance explanation.



(a) Dataset 1: High Separation

(b) Dataset 2: Moderate Separation

Figure 2: Elbow and Silhouette Analysis for both datasets.

- For Dataset-2, the choice remain unambiguous as both the "Elbow" of the inertia plot and the global max of the Silhouette Analysis ($\approx 0.51\%$) align precisely at $k = 5$.

By checking if the Silhouette peak aligns with the Elbow point (as it did in Dataset 2), we ensure that the clusters are not just cohesive, but also well-separated.

1.4 (d) Limitations of the Model Selection Procedure

The comparison of the two datasets reveals significant limitations:

- **Metric Ambiguity:** As seen in Dataset-1, validation metrics can conflict. The Elbow method focuses on internal cohesion, while Silhouette focuses on separation; deciding between them requires domain knowledge.
- **Curse of Dimensionality:** In high-dimensional spaces ($d = 7$), Euclidean distances tend to concentrate. As d increases, the ratio of the distance to the nearest neighbor versus the farthest neighbor approaches 1, which can flatten the "elbow" and make clustering results unstable.

1.5 (e) Failure Modes and Detection

k-means fails when data is non-convex or has varying densities.

- **Failure Mode:** In "Anisotropic" clusters (elongated ellipsoids), k-means will incorrectly split a single cluster into two because it minimizes the distance to a central point rather than capturing the direction of variance.
- **Detection Procedure:** A principled apporach involves:
  1. **Silhouette Profiles:** Visualizing the silhouette coefficients for every point; if a cluster has many points with negative values, the k-means boundaries are likely misaligned.
  2. **Dimensionality Reduction:** Using PCA or t-SNE to project the 7D or 9D data into 2D. If visualization shows clusters being non-spherical or "sliced" across their long axis, k-means is the wrong model. It should be discarded in favor of DBSCAN or Spectral Clustering.

2 Methodology and Implementation

The script is designed to be fully compatible with the course auto-evaluation environment.

The process begins with the execution of `env.sh` from the project root to configure the Python 3.10¡ environment and install necessary libraries (NumPy, Scikit-learn, Matplotlib). After moving into the `q1` directory, the front-end script `Q1.py` is executed. The script utilizes the `urllib` and `json` libraries to fetch datasets via an API using the student Kerberos ID prefix **aib252564**. It supports two modes: direct API fetching via a dataset number or local loading from `.npy` files. The script generates `plot.png` containing the Elbow Graph, `dataset_<dataset_num>_plot2.png` containing Elbow/Silhouette Analysis and prints the identified optimal k to `stdout`.

Listing 1: Command Sequence for Environment Setup and Execution

```
# 1. Setup environment dependencies
bash env.sh

# 2. Navigate to the question directory
cd q1

# 3. Mode 1: Run using dataset API (Student ID prefix: aib2025...)
python3 Q1.py 1

# 4. Mode 2: Run using local .npy file
python3 Q1.py path_to_dataset.npy
```